980 SOFTWARE DOCUMENTATION

P/N 966802-9901-9902

A

APPLICATION	
NEXT ASSY	USED ON
223999	8101

REVISIONS			
LTR	DESCRIPTION	DATE	APPROVED

REV																														
SHEET																														

REV STATUS OF SHEETS	REV
	SHEET

UNLESS OTHERWISE SPECIFIED DIMENSIONS ARE IN INCHES
TOLERANCES
ANGLES ±1°
3 PLACE DECIMAL ±.010
2 PLACE DECIMAL ±.02

IDENTIFYING NUMBERS SHOWN IN PARENTHESES FOR REFERENCE ONLY

INTERPRET DWG IN ACCORDANCE WITH MIL-STD-100

	DATE
DWN	
CHK	
ENGR	
QA	
APVD	
CONTR NO	
DESIGN ACTIVITY RELEASE	

TEXAS INSTRUMENTS
INCORPORATED
Equipment Group Dallas, Texas

PD,CLPCHK,132-COLUMN LINE PRINTER-PDT980

SIZE	CODE IDENT NO	DRAWING NO
A	96214	966802-9901
SCALE	D	SHEET 1 of 4

TI-9923-C

1.0 SCOPE

This test is performed to assure that the 132-column line printers (part number 966792) are functioning properly when interfaced with the Texas Instruments Model 980A Computer. The test nomenclature is CLPCHK.

2.0 RELATED MATERIALS

a. TTYEIA linkable I/O Module Object (TI Part No. 942932)

3.0 EQUIPMENT REQUIREMENTS

a. Model 980 Computer with at least 4096 words of memory.
b. Model 700/730/733 Electronic Data Terminal or ASR-33 Teletypewriter.
c. 132-column line printer (966792) properly interfaced with the Model 980 Central Processing Unit (CPU)

4.0 TEST DESCRIPTION

This test is used to determine that the line printer accepts and prints data only as specified. The test pattern consists of two parts:

a. Part one consists of 64 lines of 80 or 132 characters each. The characters increase in sort order from " " (X'20') to "←" (X'5F) from one line to the next.
b. Part two consists of 80 or 132 lines of ripple dump of the 64 characters. If the SENSE Switch 2 is ON, the ripple dump will be continued indefinitely.

5.0 OPERATING PROCEDURES

a. Load the CLPCHK program using the appropriate input device and loader.
b. Select START.
c. Enter the I/O bus group 0-3, (0 NORMAL) and external register address (50 NORMAL) as requested by the program.

d. When certain error conditions are encountered, the program prints an informative message and then idles. To try the error condition again, select START. Further messages are inhibited by setting the SENSE Switch 1 as directed by the program.

6.0 MESSAGES

a. ENTER GROUP NUMBER 0-3 (NORMAL IS 0)
Enter (via the data terminal) the number of the data bus group to which the line printer is attached.

b. ENTER REGISTER ADDRESS IN HEX (NORMAL IS 50)
Enter the external register address of the line printer.

c. LOWER CASE CHARACTER SET AVAILABLE? (Y = YES, N = NO)
Enter (via data terminal) a "Y" if both upper and lower case characters are available or enter a "N" if only upper case characters are available.

d. 80 COLUMN FORMAT? (Y = 80 COL, N = 132 COL)
Enter (via data terminal) a "Y" if 80 column paper is in printer or enter a "N" if 132 column paper is in printer.

e. PRINTER OFF-LINE
CORRECT CONDITION AND HIT START
The line printer is off-line. Check for mechanical failure or out-of-paper conditions.

f. PRINTER NOT REQUESTING DATA
SET SSW1 ON TO SUPPRESS FURTHER MESSAGES
HIT START
The line printer, having received data, is supposed to return a status, indicating that on-line/off-line status may be requested and, if found satisfactory to the unit, may be strobed to enter data into the line printer's buffer.

g. TEST COMPLETE
The program is finished with output patterns.

NOTE

All data transmission is under interrupt control; therefore, the CPU may appear to be in idle. As long as the line printer is operating, no error condition has been detected which would cause a "hard" idle state to be entered.

7.0 LINKAGE

For linkage information see - 9903 of this document.

A

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TEXAS INSTRUMENTS INCORPORATED
Equipment Group Dallas, Texas

AL,CLPCHK,132-COLUMN LINE PRINTER-PDT980

SIZE	CODE IDENT NO	DRAWING NO
A	96214	966802-9902

SCALE | REV. D | SHEET 1 OF 15

TI-9923-C

```
                 *  TITLE=CLPCHK CENTRONIC LINE PRINTER CHECK
                 *
                 *  AUTHOR=RON TATUM, ROBERT BERGLER
                 *
                 *  REVISION=05/30/75 BOB LEDBETTER
                 *   * CHANGED TO ADD COL FIX FOR 132 COL
                 *   * DELETED NULLS - NEW TTYEIA
                 *
                 *  REVISION=06/01/74 BOB LEDBETTER
                 *   * CHANGED TO RUN WITH EITHER PRINTER I/F
                 *   * CHANGED TO REMOVE UNNEED ERROR MESSAGES
                 *
                 *  REVISION=01/24/74 DON ROBERTSON
                 *   * CHANGED TO RUN ON 733
                 *
                 *  COMPUTER=980A,SAPG
                 *
                 *  ABSTRACT= THIS IS THE CENTRONICS LINE PRINTER PDT.
                 *
                        IDT  CLPCHK
                        HED DATA BUS LINE PRINTER PERFORMANCE DEMONSTRATION
                 ***********************************************************
                 *
                 *        CENTRONICS MODEL 101A/102A LINE PRINTER PERFORMANCE
                 *        DEMONSTRATION TEST WRITTEN FOR TEXAS INSTRUMENTS,
                 *        INC., AUGUST 1973, BY ROBERT M. BERGELER. THIS PDT
                 *        IS A MODIFIED VERSION OF THE PRINTEC LINE PRINTER
                 *        PDT WRITTEN BY R.H. TATUM, JUNE 1972.
                 *
                 *        THIS PROGRAM PRINTS ALL 64 VALID CHARACTERS IN ALL
                 *        132 PRINT POSITIONS, THEN FORM FEEDS, THEN PRINTS
                 *        132 LINES OF RIPPLE DUMP.
                 *
                 *        IF IT IS DESIRED TO OBTAIN MORE RIPPLE DUMP,
                 *        PLACE SENSE SWITCH 2 ON.
                 *
                 ***********************************************************
            0000 A      EQU  0
            0001 E      EQU  1
            0002 X      EQU  2
            0003 M      EQU  3
            0004 S      EQU  4
            0005 L      EQU  5
            0006 B      EQU  6
            0001 BR     EQU  1
            0007 P      EQU  7
            0008 ST     EQU  8
                        REF  WRITES
                        REF  READS
            8D8A CRLF   EQU  >8D8A
            0000 MS1    EQU  $
     0000   747A        BRL  *WRITS          WRITE HEADER ON CONSOLE
   P 0001   009F        DATA HEDLIN
```

TEXAS INSTRUMENTS
INCORPORATED
AUSTIN, TEXAS

DOCUMENT NUMBER REVISION
966802-9902 *D

DATA BUS LINE PRINTER PERFORMANCE DEMONSTRATION TEST SHEET 0003

```
  0002  7478  0054  MS1B    BRL   *WRITS              ASK FOR GROUP NUMBER
P 0003  0198  0055          DATA  D1$
  0004  7477  0056          BRL   *READ               READ GROUP NUMBER
P 0005  0084  0057          DATA  TEMP
  0006  0001  0058          DATA  1
  0007  007C  0059          LDA   TEMP                LOAD A WITH GROUP NUMBER
  0008  6730  0060          CPL   =>30                GROUP NUMBER >=0?
  0009  CDC0  0061          SLE
  000A  78F7  0062          BRU   MS1B                NO - REQUEST GROUP AGAIN
  000B  6733  0063          CPL   =>33                GROUP NUMBER <=3?
  000C  CD80  0064          SGE
  000D  78F4  0065          BRU   MS1B                NO - REQUEST GROUP AGAIN
  000E  3F03  0066          AND   =3                  MASK OF ASCII CODE
  000F  0F00  0067          LDE   =0
  0010  8875  0068          STE   SKELG               ZERO SKEL I/O COMMAND
  0011  C8C9  0069          LLA   9                   PUT GROUP IN PROPER BIT
  0012  8073  0070          STA   SKELG               POSITIONS AND SAVE
  0013  7467  0071  MS2     BRL   *WRITS              REQUEST REG ADDRESS
P 0014  01AE  0072          DATA  D2$
  0015  7466  0073          BRL   *READ               READ REG ADDRESS
P 0016  0084  0074          DATA  TEMP
  0017  0002  0075          DATA  2
  0018  006B  0076          LDA   TEMP                MOST SIG HEX CHARACTER (ASCII
  0019  086B  0077          LDE   TEMP+1              LEAST SIG HEX CHARACTER(ASCII
  001A  17FE  0078          LDX   =-2                 SET INDEX FOR 2 CHAR CONVERT
  001B  3F7F  0079  GO      AND   =>7F                INSURE 7-BIT ASCII CHARACTER
  001C  7056  0080          BRL   RANGE               IS CHARACTER A THRU F?
  001D  0041  0081          DATA  >41,>46
  001E  0046
  001F  7802  0082          BRU T09                   NO
  0020  2709  0083          ADD   =9                  YES-CONVERT TO HEX
  0021  7804  0084          BRU   MASK
  0022  7050  0085  T09     BRL   RANGE               IS CHARACTER 0 THRU 9?
  0023  0030  0086          DATA  >30,>39
  0024  0039
  0025  78ED  0087          BRU   MS2                 REQ REG ADDRESS AGAIN
  0026  3F0F  0088  MASK    AND   =>F                 LAST 4-BITS IS HEX CHARACTER
  0027  C781  0089          REX   A,E
  0028  40F2  0090          BIX   GO                  GO CONVERT OTHER CHARACTER
  0029  C8C4  0091          LLA   4                   BUILD REG ADDRESS
  002A  C481  0092          ROR   A,E
  002B  3F5F  0093          AND   =>5F                INSURE REG ADD IS 5F OR LESS
  002C  0059  0094          LDA   SKELG
  002D  C490  0095          ROR   E,A                 OR GROUP WITH ADDRESS
  002E  C503  0096          RMO   A,M
  002F  17F4  0097          LDX   =LOCS-MODINS
  0030  0A62  0098  ADLOOP  LDE   LOCS,X              PICK UP LOCATION AND
  0031  026D  0099          LDA   MODINS,X            INSTRUCTION
  0032  C480  0100          ROR   M,A                 MODIFY THE ADDRESS PORTION
  0033  C516  0101          RMO   E,B                 MOVE LOCATION TO BASE REG AND
  0034  8100  0102          STA   0,BR                STORE INSTRUCTION BASE REL
  0035  40FA  0103          BIX   ADLOOP
  0036  B000  0104          DLD   $+1                 PUT TRAP INSTRUCTION AT
```

DATA BUS LINE PRINTER PERFORMANCE DEMONSTRATION TEST

```
  0037  D8C0          @SSB DBINT1          I/O TRAP LOCATION
P 0038  00C6
  0039  A706          DST   =6
  003A  7440          BRL   *WRIT$         ASK ABOUT UPPER/LOWER CASE
P 003B  0232          DATA DC$
  003C  743F          BRL   *READ          READ Y OR N
P 003D  0084          DATA TEMP
  003E  0001          DATA 1
  003F  0044          LDA   TEMP
  0040  3F7F          AND   =>7F           7-BIT ASCII
  0041  6759          CPL   =>59           COMPARE WITH 'Y'
  0042  B03C          DLD   LCONT          LOAD UP/LOWER CHAR SET COUNTS
  0043  CD20          SEQ                  SKIP IF LOWER CASE AVAIL
  0044  B03C          DLD   UCONT          ELSE LOAD UP CASE ONLY COUNTS
  0045  A037          DST   MAXLN          SAVE COUNTS
  0046  7434          BRL   *WRIT$         ASK ABOUT 80 VS 132 COL
P 0047  01C8          DATA D3$
  0048  7433          BRL   *READ          WAIT FOR ANSWER
P 0049  0084          DATA TEMP
  004A  0001          DATA 1
  004B  0038          LDA   TEMP
  004C  3F7F          AND   =>7F           7-BIT ASCII
  004D  6759          CPL   =>59           COMPARE WITH 'Y'
  004E  CDA0          SNE
  004F  7813          BRU   COLFIX         80 COL - FIX COL FORMATS
  0050  0000          @LDA =-132           FIX FOR DUMP
  0051  FF7C
  0052  8400          @STA DMPLP1+1
P 0053  00E8
  0054  0000          @LDA =132            FIX FOR RIPPLE
  0055  0084
  0056  8400          @STA RIPPLE+2
P 0057  00F1
  0058  0000          LDA   $+1            ANOTHER FIX FOR RIPPLE
  0059  17D4          LDX   =-44
  005A  8400          @STA DMPLP4-1
P 005B  00FD
  005C  8000          DLD   $+1            FIX HEADER
  005D  7400          @BRL WT1WDS
P 005E  011A
  005F  A400          @DST HEDFIX
P 0060  00DD
  0061  D880          @LSB IDOL
P 0062  00BD
  0063  07B0   COLFIX LDA   =-80           FIX FOR DUMP
  0064  8400          @STA DMPLP1+1
P 0065  00E8
  0066  0750          LDA   =80            FIX FOR RIPPLE
  0067  8400          @STA RIPPLE+2
P 0068  00F1
  0069  0000          LDA   $+1            ANOTHER FIX FOR RIPPLE
  006A  17E5          LDX   =-27
  006B  8400          @STA DMPLP4-1
```

```
P 006C  00FD
  006D  0000  0147          LDA  $+1               FIX FOR HEADER
  006E  7801  0148          BRU  $+2
  006F  8400  0149          @STA HEDFIX
P 0070  00DD
  0071  D880  0150          @LSB IDOL
P 0072  00BD
  0073  C556  0151  RANGE   RMO  L,B               B=DATA BASE
  0074  6100  0152          CPL  0,BR              TEST LOW
  0075  CDC0  0153          SLE  A
  0076  7902  0154          BRU  2,BR              RETURN LOW
  0077  6101  0155          CPL  1,BR              TEST IN RANGE
  0078  CD80  0156          SGE  A
  0079  7899  0157          BRU  MS2               ERROR ASK AGAIN
  007A  7903  0158          BRU  3,BR              RETURN IN RANGE
        0000  0159  WRITS   DATA WRITES
X 007B  0000
        0001  0160  READ    DATA READS
X 007C  0000
  007D  0000  0161  MAXLN   DATA 0,0
  007E  0000
  007F  005F  0162  LCONT   DATA >5F,>7D
  0080  007D
  0081  0040  0163  UCONT   DATA 64,>5F
  0082  005F
  0083  0000  0164  FSTCHR  DATA 0
  0084  0000  0165  TEMP    DATA 0,0
  0085  0000
  0086  0000  0166  SKELG   DATA 0
P 0087  00CA  0167          DATA WDS1
P 0088  00CE  0168          DATA WDS2
P 0089  0125  0169          DATA WT1WT4   A WDS1 INST
P 008A  012B  0170          DATA WT1      AN RDS 0 INST
P 008B  012F  0171          DATA WT1RD2   AN RDS 1 INST
P 008C  0134  0172          DATA WT1WT2   A WDS 2
P 008D  0138  0173          DATA WT1WT3   A WDS 2
P 008E  017D  0174          DATA PUT3SB   A WDS 1
P 008F  0183  0175          DATA PUT3SC   AN RDS 0
P 0090  0187  0176          DATA PUT3SD   AN RDS 1
P 0091  018C  0177          DATA PUT3SE   A WDS 2
P 0092  0190  0178          DATA PUT3SF   A WDS 2
        0093  0179  LOCS    EQU  $
  0093  D820  0180          WDS  0
  0094  D820  0181          WDS  0
  0095  D821  0182          WDS  1
  0096  D800  0183          RDS  0
  0097  D801  0184          RDS  1
  0098  D822  0185          WDS  2
  0099  D822  0186          WDS  2
  009A  D821  0187          WDS  1
  009B  D800  0188          RDS  0
  009C  D801  0189          RDS  1
  009D  D822  0190          WDS  2
```

```
  009E  D822  0191         WDS   2
        009F  0192  MODINS EQU   $
  009F  8D8A  0193  HEDLIN DATA  CRLF
  00A0  C4C1  0194  HED    DATA  'DATA BUS LINE PRINTER PERFORMANCE '
  00A1  D4C1
  00A2  A0C2
  00A3  D5D3
  00A4  A0CC
  00A5  C9CE
  00A6  C5A0
  00A7  D0D2
  00A8  C9CE
  00A9  D4C5
  00AA  D2A0
  00AB  D0C5
  00AC  D2C6
  00AD  CFD2
  00AE  CDC1
  00AF  CEC3
  00B0  C5A0
  00B1  C4C5  0195         DATA  'DEMONSTRATION TEST. '
  00B2  CDCF
  00B3  CED3
  00B4  D4D2
  00B5  C1D4
  00B6  C9CF
  00B7  CEA0
  00B8  D4C5
  00B9  D3D4
  00BA  AEA0
  00BB  8D8A  0196         DATA  CRLF,0
  00BC  0000
P 00BD  00C8  0197  IDOL   DATA  DBINT1+2
  00BE  0100  0198         DATA  >0100
  00BF  0000  0199  SVFILE DATA  0,0,0,0,0,0,0
  00C0  0000
  00C1  0000
  00C2  0000
  00C3  0000
  00C4  0000
  00C5  0000
  00C6  0000  0200  DBINT1 DATA  0,>0100
  00C7  0100
  00C8  0800  0201         @LDE  =>8000          DISCONNECT LINE PRINTER
  00C9  8000
  00CA  D820  0202  WDS1   WDS   0
  00CB  0001  0203         DATA  1
  00CC  0800  0204         @LDE  =>4000          ENABLE OUTPUT INTERRUPTS
  00CD  4000
  00CE  D820  0205  WDS2   WDS   0
  00CF  0001  0206         DATA  1
  00D0  707F  0207         BRL   EJECT           SKIP TO TOP OF FORM
  00D1  17F2  0208         LDX   =-14            SPACE OVER TO CENTER HEADING
```

```
  00D2  1F20          LDM   =>20        LOAD M WITH SPACE CODE
  00D3  7046   SPCLP  BRL   WT1WDS      M TO LINE PRINTER
  00D4  40FE          BIX   SPCLP
  00D5  0F01          LDE   =1          E DETERMINES WHICH HALF TO OU
  00D6  02C9   HEDPLP LDA   HED,X       PICK UP PAIR OF HEADER CHARS
  00D7  CC80          SNZ   A           ZERO'S ENDS IT
  00D8  780A          BRU   DMP1
  00D9  CA08   HDLP1  CRA   8           GET PROPER CHARACTER
  00DA  C783          REX   A,M
  00DB  703E          BRL   WT1WDS      M TO LINE PRINTER
  00DC  C783          REX   A,M         PRINT BLANK - M STILL HAS IT
  00DD  7400   HEDFIX *BRL  WT1WDS      80 COL - BRU $+2
P 00DE  011A
  00DF  C311          RIN   E,E
  00E0  CC41          SOD   E           E ODD - CHAR PAIR COMPLETE
  00E1  78F7          BRU   HDLP1
  00E2  40F3          BIX   HEDPLP      GO GET ANOTHER PAIR
  00E3  702A   DMP1   BRL   SPACE       CARRAGE RTN LINE FEED
  00E4  0098          LDA   MAXLN       NUMBER OF CHARACTERS AVAIL
  00E5  8025          STA   LINCNT      PUT NUMBER OF CHAR IN LINCNT
  00E6  1F20          LDM   =>20        START WITH SPACES
  00E7  1000   DMPLP1 *LDX  =-132       80 COL - *LDA  =-80
  00E8  FF7C
  00E9  7030   DMPLP2 BRL   WT1WDS      M TO LINE PRINTER
  00EA  40FE          BIX   DMPLP2      FILL LINE
  00EB  7022          BRL   SPACE       CARRAGE RTN LINE FEED
  00EC  C333          RIN   M,M         NEXT CHARACTER
  00ED  481D          DMT   LINCNT      ALL DONE?
  00EE  78F8          BRU   DMPLP1      NO, NOT JUST YET
  00EF  7060   RIPPLE BRL   EJECT       SKIP TO TOP OF FORM
  00F0  0000          *LDA  =132        80 COL - *LDA =80  LINE COUNT
  00F1  0084
  00F2  8018          STA   LINCNT
  00F3  071E   DMPLP6 LDA   =>1E        SET UP FIRST CHARACTER
  00F4  808E          STA   FSTCHR
  00F5  508D   DMPLP3 IMO   FSTCHR      FIRST CHARACTER - 1
  00F6  008C          LDA   FSTCHR
  00F7  6400          *CPL  MAXLN+1     IS FSTCHR > MAX CHARACTER
P 00F8  007E
  00F9  CD40          SGT
  00FA  78F8          BRU   DMPLP6      YES
  00FB  8044          STA   LASCHR      NO
  00FC  7060          BRL   GENASC      GENERATE FIRST TRIAD FOR LINE
  00FD  17D4          LDX   =-44        80 COL - LDX  =-27
  00FE  7073   DMPLP4 BRL   PUT3        PUT OUT I-TH TRIAD TO LINE
  00FF  705D          BRL   GENASC      GENERATE ANOTHER TRIAD
  0100  40FD          BIX   DMPLP4
  0101  700C          BRL   SPACE       CARRAGE RTN LINE FEED
  0102  4808          DMT   LINCNT      DONE YET?
  0103  78F1          BRU   DMPLP3      NO,SO GO ON RIPPLING
  0104  CC94          SSN   4           CONTINUE RIPPLING
  0105  78E9          BRU   RIPPLE      YES, GO TO IT
  0106  7405          BRL   *WRITE      PRINT TEST COMPLETE
```

```
P 0107  0228  0258         DATA DBS
  0108  CE01  0259         IDL  1
  0109  7C00  0260         @BRU MS1B
P 010A  0002
  010B  0000  0261  LINCNT DATA 0
        0000  0262  WRITE  DATA WRITE$
X 010C  0000
  010D  0000  0263  RETRY  DATA 0
  010E  C550  0264  SPACE  RMO  L,A              STORE RETURN ADDRESS
  010F  8009  0265         STA  SPACER          IN SPACER
  0110  C7C3  0266         REX  S,M             SAVE M
  0111  17FE  0267         LDX  =-2
  0112  1A06  0268  SPCLP1 LDM  SPACES,X         LOAD CR/LF
  0113  7006  0269         BRL  WT1WDS          M TO LINE PRINTER
  0114  40FD  0270         BIX  SPCLP1
  0115  C7C3  0271         REX  S,M             RESTORE M
  0116  7C02  0272         BRU  *SPACER         RETURN TO CALLER
  0117  008D  0273         DATA >8D,>8A
  0118  008A
        0119  0274  SPACES EQU  $
  0119  0000  0275  SPACER DATA 0
  011A  D8E0  0276  WT1WDS @SRF SVFILE
P 011B  008F
  011C  0022  0277         LDA  WT1WDS  GET RETRY ADDRESS
  011D  80EF  0278         STA  RETRY
  011E  B000  0279         DLD  $+1
  011F  D8C0  0280         @SSB STSV1
P 0120  0129
  0121  A706  0281         DST  =6
  0122  C530  0282         RMO  M,A
  0123  3F7F  0283         AND  =>7F
  0124  C200  0284         RIV  A,A
  0125  D821  0285  WT1WT4 WDS  1       WRITE OUT CHARACTER
  0126  0080  0286         DATA >80
  0127  78FD  0287         BRU  WT1WT4  REPEAT UNTIL DEVICE ACCEPTS
  0128  CE00  0288         IDL  0       WAIT FOR DATA BUS INTERRUPT
  0129  0000  0289  STSV1  DATA 0,0     STATUS, PC GET STORED HERE
  012A  0000
  012B  D800  0290  WT1    RDS  0       READ STATUS TO A-REG WITH BUSY BIT ON
  012C  0000  0291         DATA 0
  012D  DB16  0292         TABO 6       CHECK FOR DESIRED STATUS
  012E  7812  0293         BRU  BADST2  PRINTER NOT REQUESTING DATA - ERROR
  012F  D801  0294  WT1RD2 RDS  1       READ DATA TO CHECK PRINTER ON-LINE
  0130  0000  0295         DATA 0
  0131  DB11  0296         TABO 1       CHECK EXPECTED DATA - INDICATES LP READ
  0132  7814  0297         BRU  POFLIN  PRINTER UNEXPECTEDLY OFF-LINE
  0133  0F07  0298         LDE  =7
  0134  D822  0299  WT1WT2 WDS  2       WRITE BIT TO STROBE UNIT
  0135  0001  0300         DATA 1
  0136  CA10  0301         CRA  16      WAIT APPROX. 4 MICROSECONDS
  0137  0F17  0302         LDE  =>17
  0138  D822  0303  WT1WT3 WDS  2
  0139  0001  0304         DATA 1
```

DATA BUS LINE PRINTER PERFORMANCE DEMONSTRATION TEST SHEET 0009

```
  013A  00EF  0305         LDA   STSV1+1  FETCH STORED STATUS REGISTER
  013B  C508  0306         RMO   A,ST     RESTORE IT
  013C  D8A0  0307         @LRF  SVFILE
P 013D  00BF
  013E  C557  0308         RMO   L,P      RETURN TO CALLER
P 013F  011A  0309  WT1WDS DATA  WT1WDS
  0140  0022  0310  LASCHR DATA  >0022    START BY TRYING TO GEN 4TH CHARACTER
  0141  CC98  0311  BADST2 SSN   8        PRINTER NOT REQUESTING DATA IF HERE
  0142  7807  0312         BRU   ERHND
  0143  74C8  0313         BRL   *WRITE
P 0144  01FA  0314         DATA  D8$
  0145  CE00  0315         IDL
  0146  7803  0316         BRU   ERHND
  0147  74C4  0317  POFLIN BRL   *WRITE
P 0148  01DE  0318         DATA  D6$
  0149  CE00  0319         IDL
  014A  0000  0320  ERHND  @LDA  =>100
  014B  0100
  014C  C508  0321         RMO   A,ST
  014D  D8A0  0322         @LRF  SVFILE
P 014E  00BF
  014F  7CBD  0323         BRU   *RETRY
  0150  17FD  0324  EJECT  LDX   =-3      WRITE FORM FEED,CR,LF USING WT1WDS
  0151  C550  0325         RMO   L,A
  0152  8009  0326         STA   EJECTR
  0153  C7C3  0327         REX   S,M
  0154  1A07  0328  EJECT1 LDM   CONCAR,X
  0155  70C4  0329         BRL   WT1WDS
  0156  40FD  0330         BIX   EJECT1   GO PUT OUT NEXT CONTROL CHARACTER
  0157  C7C3  0331         REX   S,M
  0158  7C03  0332         BRU   *EJECTR
  0159  000D  0333         DATA  >000D,>000C,>000A
  015A  000C
  015B  000A
        015C  0334  CONCAR EQU   $
  015C  0000  0335  EJECTR DATA  0
  015D  800F  0336  GENASC STA   GENASA
  015E  900E  0337         STX   GENASX
  015F  17FD  0338         LDX   =-3
  0160  00DF  0339         LDA   LASCHR
  0161  6400  0340  GENALP @CPL  MAXLN+1
P 0162  007E
  0163  CD40  0341         SGT            IF NOT, GENERATE NEXT
  0164  071F  0342         LDA   =>1F     RESET CHAR. GENERATED TO SPACE
  0165  C300  0343         RIN   A,A      NEXT CHARACTER
  0166  820A  0344         STA   TRIAD,X
  0167  40F9  0345         BIX   GENALP
  0168  80D7  0346         STA   LASCHR   AND SAVE FOR NEXT TIME THROUGH
  0169  0002  0347         LDA   GENASA
  016A  1002  0348         LDX   GENASX
  016B  C557  0349         RMO   L,P      RETURN
  016C  0000  0350  GENASA DATA  0
  016D  0000  0351  GENASX DATA  0
```

```
  016E  0020  0352          DATA >20,>21,>22
  016F  0021
  0170  0022
        0171  0353  TRIAD   EQU  $
P 0171  0172  0354  PUT3$   DATA PUT3
  0172  D8E0  0355  PUT3    @SRF SVFILE   SAVE USEFUL REGS
P 0173  00BF
  0174  B000  0356          DLD  $+1
  0175  D8C0  0357          @SSB PUT3I
P 0176  0181
  0177  A706  0358          DST  =6
  0178  00F8  0359          LDA  PUT3$
  0179  8093  0360          STA  RETRY
  017A  17FD  0361          LDX  =-3      SET TO WRITE 3 CHARS
  017B  02F5  0362  PUT3SA  LDA  TRIAD,X  GET I-TH CHARACTER
  017C  C200  0363          RIV  A,A
  017D  D821  0364  PUT3SB  WDS  1        LOAD CHARACTER TO DATA MODULE
  017E  0080  0365          DATA >80
  017F  78FD  0366          BRU  PUT3SB   WAIT FOR DEVICE TO ACCEPT CHAR'.
  0180  CE02  0367          IDL  2        WAIT FOR INTERRUPT
  0181  0000  0368  PUT3I   DATA 0,0
  0182  0000
  0183  D800  0369  PUT3SC  RDS  0        ISSUE READ STATUS
  0184  0000  0370          DATA 0
  0185  DB16  0371          TABO 6        CHECK FOR DESIRED STATUS - OA = REQ'. D
  0186  78BA  0372          BRU  BADST2   BAD STATUS RECEIVED
  0187  D801  0373  PUT3SD  RDS  1        READ DATA TO CHECK PTR ON LINE
  0188  0000  0374          DATA 0
  0189  DB11  0375          TABO 1        CHECK EXPECTED DATA - INDICATES LP REA
  018A  78BC  0376          BRU  POFLIN   PRINTER IS OFF-LINE, SHOULDN'T BE
  018B  0F07  0377          LDE  =7
  018C  D822  0378  PUT3SE  WDS  2
  018D  0001  0379          DATA 1
  018E  CA10  0380          CRA  16       WAIT 4 MICS OR SO
  018F  0F17  0381          LDE  =>17
  0190  D822  0382  PUT3SF  WDS  2
  0191  0001  0383          DATA 1
  0192  00EF  0384          LDA  PUT3I+1
  0193  C508  0385          RMO  A,ST
  0194  40E6  0386          BIX  PUT3SA   GO PUT OUT NEXT CHAR
  0195  D8A0  0387          @LRF SVFILE   RESTORE
P 0196  00BF
  0197  C557  0388          RMO  L,P      AND WE'VE PRINTED A TRIAD
  0198  8D8A  0389  D1$     DATA CRLF
  0199  C5CE  0390          DATA 'ENTER GROUP NUMBER 0-3 (NORMAL IS 0).'
  019A  D4C5
  019B  D2A0
  019C  C7D2
  019D  CFD5
  019E  D0A0
  019F  CED5
  01A0  CDC2
  01A1  C5D2
```

DATA BUS LINE PRINTER PERFORMANCE DEMONSTRATION TEST   SHEET 0011

```
01A2  A0B0
01A3  ADB3
01A4  A0A8
01A5  CECF
01A6  D2CD
01A7  C1CC
01A8  A0C9
01A9  D3A0
01AA  B0A9
01AB  AEFF
01AC  8D8A  0391        DATA CRLF,0
01AD  0000
01AE  8D8A  0392  D2$   DATA CRLF
01AF  C5CE  0393        DATA 'ENTER REGISTER ADDRESS IN HEX (NORMAL IS 50).'
01B0  D4C5
01B1  D2A0
01B2  D2C5
01B3  C7C9
01B4  D3D4
01B5  C5D2
01B6  A0C1
01B7  C4C4
01B8  D2C5
01B9  D3D3
01BA  A0C9
01BB  CEA0
01BC  C8C5
01BD  D8A0
01BE  A8CE
01BF  CFD2
01C0  CDC1
01C1  CCA0
01C2  C9D3
01C3  A0B5
01C4  B0A9
01C5  AEFF
01C6  8D8A  0394        DATA CRLF,0
01C7  0000
01C8  8D8A  0395  D3$   DATA CRLF
01C9  B8B0  0396        DATA '80 COLUMN FORMAT?(Y=80 COL N=132 COL).'
01CA  A0C3
01CB  CFCC
01CC  D5CD
01CD  CEA0
01CE  C6CF
01CF  D2CD
01D0  C1D4
01D1  BFA8
01D2  D9BD
01D3  B8B0
01D4  A0C3
01D5  CFCC
01D6  A0CE
```

```
01D7  8DB1
01D8  83B2
01D9  A0C3
01DA  CFCC
01DB  A9AE
01DC  8D8A  0397        DATA CRLF,0
01DD  0000
01DE  8D8A  0398  D6S   DATA CRLF,'PRINTER OFF-LINE.',CRLF
01DF  D0D2
01E0  C9CE
01E1  D4C5
01E2  D2A0
01E3  CFC6
01E4  C6AD
01E5  CCC9
01E6  CEC5
01E7  AEFF
01E8  8D8A
01E9  C3CF  0399         DATA 'CORRECT CONDITION & HIT START.'
01EA  D2D2
01EB  C5C3
01EC  D4A0
01ED  C3CF
01EE  CEC4
01EF  C9D4
01F0  C9CF
01F1  CEA0
01F2  A6A0
01F3  C8C9
01F4  D4A0
01F5  D3D4
01F6  C1D2
01F7  D4AE
01F8  8D8A  0400        DATA CRLF,0
01F9  0000
01FA  8D8A  0401  D8S   DATA CRLF
01FB  D0D2  0402        DATA 'PRINTER NOT REQUESTING DATA.'
01FC  C9CE
01FD  D4C5
01FE  D2A0
01FF  CECF
0200  D4A0
0201  D2C5
0202  D1D5
0203  C5D3
0204  D4C9
0205  CEC7
0206  A0C4
0207  C1D4
0208  C1AE
0209  8D8A  0403        DATA CRLF
020A  D4D5  0404        DATA 'TURN ON SSW 1 TO SUPPRESS FURTHER MESSAGES.'
020B  D2CE
```

```
020C   A0CF
020D   CEA0
020E   D3D3
020F   D7A0
0210   B1A0
0211   D4CF
0212   A0D3
0213   D5D0
0214   D0D2
0215   C5D3
0216   D3A0
0217   C6D5
0218   D2D4
0219   C8C5
021A   D2A0
021B   CDC5
021C   D3D3
021D   C1C7
021E   C5D3
021F   AEFF
0220   8D8A   0405             DATA CRLF,'HIT START.',CRLF,0
0221   C8C9
0222   D4A0
0223   D3D4
0224   C1D2
0225   D4AE
0226   8D8A
0227   0000
0228   8D8A   0406   DBS       DATA CRLF,'TEST COMPLETE.',CRLF,0
0229   D4C5
022A   D3D4
022B   A0C3
022C   CFCD
022D   D0CC
022E   C5D4
022F   C5AE
0230   8D8A
0231   0000
0232   8D8A   0407   DCS       DATA CRLF
0233   CCCF   0408             DATA 'LOWER CASE CHARACTER SET AVAILABLE?(Y=YES '
0234   D7C5
0235   D2A0
0236   C3C1
0237   D3C5
0238   A0C3
0239   C8C1
023A   D2C1
023B   C3D4
023C   C5D2
023D   A0D3
023E   C5D4
023F   A0C1
0240   D6C1
```

```
0241   C9CC
0242   C1C2
0243   CCC5
0244   BFA8
0245   D9BD
0246   D9C5
0247   D3A0
0248   CEBD   0409          DATA 'N=NO)',CRLF,0
0249   CECF
024A   A9FF
024B   8D8A
024C   0000
       0000   0410          END  MS1
```

TEXAS INSTRUMENTS
INCORPORATED
AUSTIN, TEXAS

DOCUMENT NUMBER REVISION

966802-9902 *D

DATA BUS LINE PRINTER PERFORMANCE DEMONSTRATION TEST SHEET 0015

A	0000	ADLOOP	0030	B	0006	BADST2	0141
BR	0001	COLFIX	0063	CONCAR	015C	CRLF	8D8A
D1S	0198	D2S	01AE	D3S	01C8	D6S	01DE
D8S	01FA	DBS	0228	DBINT1	00C6	DCS	0232
DMP1	00E3	DMPLP1	00E7	DMPLP2	00E9	DMPLP3	00F5
DMPLP4	00FE	DMPLP6	00F3	E	0001	EJECT	0150
EJECT1	0154	EJECTR	015C	ERHND	014A	FSTCHR	0083
GENALP	0161	GENASA	016C	GENASC	015D	GENASX	016D
GO	0018	HDLP1	00D9	HED	00A0	HEDFIX	00DD
HEDLIN	009F	HEDPLP	00D6	IDOL	00BD	L	0005
LASCHR	0140	LCONT	007F	LINCNT	010B	LOCS	0093
M	0003	MASK	0026	MAXLN	007D	MODINS	009F
MS1	0000	MS1B	0002	MS2	0013	P	0007
POFLIN	0147	PUT3	0172	PUT3S	0171	PUT3I	0181
PUT3SA	0178	PUT3SB	017D	PUT3SC	0183	PUT3SD	0187
PUT3SE	018C	PUT3SF	0190	RANGE	0073	READ	007C
READS	0001	RETRY	010D	RIPPLE	00EF	S	0004
SKELG	0086	SPACE	010E	SPACER	0119	SPACES	0119
SPCLP	00D3	SPCLP1	0112	ST	0008	STSV1	0129
SVFILE	00BF	T09	0022	TEMP	0084	TRIAD	0171
UCONT	0081	WDS1	00CA	WDS2	00CE	WRITS	007B
WRITE	010C	WRITES	0000	WT1	012B	WT1RD2	012F
WT1WDS	013F	WT1WDS	011A	WT1WT2	0134	WT1WT3	0138
WT1WT4	0125	X	0002				

0000 ERRORS

A

APPLICATION	
NEXT ASSY	USED ON
223999	8101

REVISIONS

LTR	DESCRIPTION	DATE	APPROVED

REV

SHEET

REV STATUS OF SHEETS

REV

SHEET

UNLESS OTHERWISE SPECIFIED
DIMENSIONS ARE IN INCHES
TOLERANCES
ANGLES ±1°
3 PLACE DECIMAL ±.010
2 PLACE DECIMAL ±.02

IDENTIFYING NUMBERS SHOWN IN PARENTHESES FOR REFERENCE ONLY

INTERPRET DWG IN ACCORDANCE WITH MIL-STD-100

DWN DATE

CHK

ENGR

QA

APVD

CONTR NO

DESIGN ACTIVITY RELEASE

TEXAS INSTRUMENTS
INCORPORATED
Equipment Group Dallas, Texas

LML,CLPCHK,132-COL LINE PRINTER-PDT980

SIZE	CODE IDENT NO	DRAWING NO
A	96214	966802-9903

SCALE REV. D SHEET 1 of 5

TI-0923-C

```
THESE ARE THE CONTROL RECORDS AS READ.
SS,CLPCHK,CLPCHK,1,4
/*
```

```
INTERPRETATION

SS,CLPCHK,CLPCHK,1,4,0,0,0,A
/*
```

LINK EDITOR LOAD MAP FOR CLPCHK   EP= 0000   LN= 03B2   PAGE 0001

DEFINED ENTRY POINTS

DEF	LOCATION	PROGRAM	ADDRESS
ASDEC$	02E8	TTYEIA	024D
BIASC$	024D		
ASBIN$	02EC		
BIDEC$	034D		
RECON$	0360		
CONWR$	0397		
WAIT$	03A7		
READ$	0264		
WRITE$	0267		

EXTERNAL REFERENCES

REF	LOCATION	PROGRAM	ADDRESS
WRITES	0267	CLPCHK	0000
READS	0264		

TEXAS INSTRUMENTS INCORPORATED

DATE 09/03/75

LIST OF MATERIAL

LM 966802-9903 REV. D

PRINT ITEM NUMBER	QUANTITY PER ASSEMBLY	UNIT OF ISSUE	DWG. SIZE	PART NUMBER	DESCRIPTION	VENDOR PART NUMBER
0001	REF	EA		966802-1202	CDO, CLPCHK,132-COL LINE PRINTER-PDT980	
0002	REF	EA		942932-1201	CDO, TTYEIA,TERMINAL I/O MODULE-SYS980	

DRAFTSMAN	DATE	CKD. DRAFTSMAN	DATE	DESIGN ENGINEER	DATE	TITLE
Martin Sullivant	9/5/75	Martin Sullivant	9/5/75	M A Mote	9-5-75	LML, CLPCHK,132-COL LINE PRINTER-PDT980
MFG.	DATE	APPD. PROJECT ENGINEER	DATE	RELEASED	DATE	PROJECT NO.
		M A Mote	9-5-75	[illegible]	11/9/61	[illegible]

PART NUMBER LM 966802-9903 REV D